SuperStore Analytics Dashboard Documentation

1. Project Overview

This project focuses on Superstore analysis to evaluate retail business performance by analyzing sales trends, maximizing profitability, tracking regional dynamics, and understanding customer purchasing behavior.

The analysis answers key questions such as:

1. What is the overall sales performance of the company?
2. Which regions generate the highest sales and profit?
3. Which customer segments contribute the most revenue?
4. Which product categories perform best?
5. Which products are the most profitable?
6. What trends can be observed over time?

2. Data Description

The dataset contains store information such as:

Order Id, Order date, ship date, ship mode, customer id, customer name, segment, country, city, state, postal code, region, product id, category, sub category, product name, sales, quantity, discount, profit.

3. Data Preparation, Date Table & Measures

To enable analysis, we created several **calculated measures** in Power BI:

- ✓ Total Sales = Sum of the sales
- ✓ Total Profit = Sum of profit
- ✓ Total Orders = Distinct count of the order id
- ✓ Average Sales = Average of sales
- ✓ Profit Margin = (Profit/Sales) x 100
- ✓ Quantity Sold = Count of quantity
- ✓ Date Table = Year, Month Name, Day, Quarter, Month Number

4. Dashboard design & visuals

- **KPI's:** Total Sales, Total Profit, Total Orders, Average Sales, Profit Margin
- **Charts:**
  - Total sales & total profit by region
  - Total sales by segment
  - Total sales by month name
  - Total profit by product name
  - Total profit by category

    - Total profit by ship mode
    - Total sales by state
- Slicers/Filters: Year, Month name, Region, Segment, Ship mode
- Table/Matrix: Product name, Total sales, Total profit, Quantity, Profit margin, Total orders

5. Key Business Insights

- **Regional performance:** The West region generated the highest sales and profit, with $725.46K in sales and $108.42K in profit, followed by the East region, with $678.78K in sales and $91.52K in profit.
- **Customer segment performance:** The Consumer segment generated the highest revenue of $1.16M, accounting for 50.56% of total revenue.
- **Category performance**: Technology is the most profitable category, with a 17% profit margin and $145.45K in profit.
- **Product performance:** The Canon imageCLASS 2200 Advanced Copier is the most profitable product, with a 41% profit margin, $61.60K in sales, and $25.20K in profit. It is followed by the Fellowes PB500 Electric Punch Plastic Comb Binding Machine, with a 28% profit margin, $27.45K in sales, and $7.25K in profit.
- **Sales trends:** November recorded the highest sales, generating $352.46K in revenue.

6. Recommendations

- **Improve January sales performance:** Management should introduce targeted promotions, discounts on selected products, and marketing campaigns in January to attract more customers and increase order volume.
- **Review Furniture pricing and costs:** Management should investigate the reasons for the low profit margin in the Furniture category and review product pricing, supplier costs, and discount policies to improve profitability.
- **Improve Central region profitability:** Management should analyze product-level profit margins, operating costs, and discount rates in the Central region. The focus should be on promoting high-margin products and reducing unnecessary costs rather than simply increasing sales.
- **Prioritize high-performing products:** Management should ensure adequate stock of the Canon imageCLASS 2200 Advanced Copier and evaluate opportunities to increase its sales while maintaining its profit margin.
- **Increase investment in Technology:** Management should consider expanding its Technology product offerings, improving product availability, and promoting high-margin products to increase profits.

- **Plan ahead for November demand:** Management should prepare for the December sales peak by optimizing inventory levels, planning marketing campaigns early, and ensuring that popular products are available to meet customer demand.

7. Business risks

- **Low January sales:** January recorded the lowest revenue of $94.92K, with a total order of 178 orders. Management should investigate the reasons for the low sales and develop strategies to improve performance during this period.
- **Low furniture profitability:** The Furniture category generated $18.45K in profit, with a profit margin of just 2%. This indicates that the category may be facing high costs, excessive discounts, or pricing challenges.
- **Low Central region profitability:** The Central region has a profit margin of 8% despite generating substantial revenue. Management should investigate whether low margin products, high operating costs, or excessive discounts are affecting profitability.

8. Business opportunities

Focus on high-profit products: Management should focus more on the Canon image CLASS 2200 Advanced Copier because of its high profit margin and contribution to overall profit.

Invest in the Technology category: Management should invest more in the Technology category because it has the highest profit margin of 17%, creating an opportunity to improve overall profitability.

Expand successful regional strategies: Management should study the strategies used in the West region, which generated the highest sales and profit, and apply similar approaches in other regions to improve performance.